Jaejin Cha

AI Systems | Cloud & Data Platforms | Robotics & Autonomous Systems

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Summary

MS Computer Science student at Texas A&M, graduating December 2026, and Software Engineer at HPE building enterprise AI applications for natural-language data exploration, RAG, tool use, and database-backed workflows on Oracle and OCI. Research spans multi-robot communication and graph-based planning, autonomous-driving simulation, reinforcement and imitation learning, and clinical data modeling.

Technical Skills

Languages: Python, C++, SQL, PL/SQL, JavaScript; **AI/ML:** PyTorch, scikit-learn, OpenCV, reinforcement learning, imitation learning
LLM: RAG, tool calling, text-to-SQL, LangChain; **Systems:** Linux, Docker, OCI, Oracle Database 19c/26ai, Oracle APEX, REST APIs, Git

Experience

Hewlett Packard Enterprise — **Spring, TX**
Software Engineer – AI Applications and Enterprise Systems — Intern: May 2025 – May 2026 | Full-time: Jun 2026 – Present

- Build internal AI applications enabling operations leaders to query operational data in natural language and generate on-demand insights.
- Implement text-to-SQL and tool-calling workflows with Oracle AI Database, PL/SQL packages, and business-rule guardrails.
- Develop Oracle APEX interfaces with persistent chat history, session management, and OCI-integrated AI workflows.
- Built a custom vector store and similarity-search layer in Oracle 19c using PL/SQL, REST APIs, and embedding models.

NIH AIM-AHEAD Collaborative Research — **Remote**
Research Assistant - Texas A&M, University of Washington, and Emory University — Aug 2025 – Present

- Built cardiovascular EHR data pipelines for cohort construction, feature engineering, longitudinal data preparation, and survival modeling.
- Evaluated survival models using calibration, performance metrics, and subgroup fairness analysis on real-world clinical data.

ENDEAVR Institute — **College Station, TX**
Machine Learning Engineer Intern — Oct 2024 – Feb 2025

- Developed autonomous-driving imitation-learning components using GAIL and PPO, including generator-side implementation and simulation-oriented training workflows.

University of Delaware — **Newark, DE**
Summer Research Intern — Jun 2023 – Aug 2023

- Implemented DQN and DDPG models in an autonomous-driving simulator for trajectory control and energy-efficiency experiments.

Texas A&M University — **College Station, TX**
Undergraduate Research Assistant — Sep 2022 – May 2023

- Configured Apache Spark and HDFS environments on Linux clusters and developed Python/Bash tooling to analyze garbage-collection logs for runtime and memory profiling.
- Modified Peregrine source code to convert multi-label graph datasets into binary format for graph-processing workflows.

Research & Publications

Jaejin Cha, Dylan Shell. *"Planning and Execution for Multi-Robot Cyclic Rendezvous via Graphs of Convex Sets"*. IROS (accepted). — 2026

- Developed graph-based planning and execution methods for communication-aware coordination among robots following cyclic paths.

Jaejin Cha. *"Rendezvous-Based Multi-Robot Communication Under Cyclic Paths"*. Master's thesis (defended). — 2026

Selected Projects

NatureNet | Python, PyTorch, OpenCV, Flask, React.js, Node.js, Express.js — 2024

- Built a full-stack wildlife detection and alerting system using a custom CV model, inference API, and SMS/email notification workflows.

JAL-AM Trading Simulator | Python, PyTorch, NumPy — 2024 – 2025

- Formulated a market simulator as an MDP and implemented Joint-Action Learning with Agent Modeling for decision-making experiments.

Education

Texas A&M University — **College Station, TX**
M.S. Computer Science, GPA: 3.71/4.00 - Thesis in Multi-Robot Systems (Advisor: Dr. Dylan Shell) — Expected Dec 2026

Texas A&M University — **College Station, TX**
B.S. Computer Science, Minor in Mathematics — Aug 2024

Leadership & Activities

Texas Aimbots | *Computer Vision Developer* — Sep 2023 – May 2024

- Implemented OpenCV-based target detection and contour extraction for RoboMaster University League competition robots.

Aggie Coding Club | *Project Manager* — Feb 2022 – May 2022

- Led 20+ members in web-development workshops and coordinated React projects including personal websites and e-commerce interfaces.